Q1:(Please write your answers in a table only; otherwise, your answers will be neglected) (22 marks)

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	D	B	B	A	C	A	A	B	A
Question	11	12	13	14	15					
Answer	B	A	A	B	C					

+21

1. We cannot send the digital signal directly to the channel when the channel is:

   A. bandpass
   B. Bypass
   C. baseband
   D. Low pass

2. The transmission impairment refers to a signal with high energy in a very short time is

   A. Thermal Noise
   B. Induced Noise
   C. Cross-talk
   D. Impulse Noise

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3. A digital signal is a composite analog signal with

   A. finite bandwidth
   B. Infinite bandwidth
   C. zero bandwidth
   D. None of the above

4. In data communications, the non-periodic signals

   A. Sine wave
   B. Digital Signals
   C. Analog Signals with T period
   D. None of the above

5. A communication pathway that transfers data from one point to another is called

   A. Link
   B. Node
   C. Medium
   D. Topology

6. In a bus topology, the device linking all the nodes in a network is

A. Router

B. DropLine

C. Backbone

D. hub

7. Which topology covers security, robust, and eliminates the traffic factor?

A. Mesh

B. Ring

C. Star

D. Bus

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8. The bus, ring, and star topologies are mostly used in the

A. LAN

B. MAN

C. WAN

D. Internetwork

9.The repeater is used in

A. Mesh Topology

B. Ring Topology

C. Bus Topology

D. Star Topology

10. In a human voice:

A. The number of frequencies is infinite, but the range is limited

B. The number of frequencies is finite, but the range is limited

C. The number of frequencies is infinite, but the range is unlimited

11. An amplifier has an output of 20 w. What is its output in dBw:

A. 13 dBmw B. 13 dBw C. 43 dBw D. 43 dBmw

dB = 20mw

dBs = 10 Log10 20×10^3

12. The ideal channel has the value of SNR equal to:

A. ∞ B. 0 C. 1 D. ½

$\frac{1}{0} = \infty$

13. The domain of duty of the top 3 layers in TCP/IP is:

A. internet B. link C. message

14. Connecting your cellular phone to the router at home makes a network, the phone is:

A. hub B. host C. switch

15. Bit length is measured by:

A. bits B. bits/s C. meters

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Q2: Suppose that a digitized TV picture is to be transmitted from a source that uses a 480×500 matrix of picture elements (pixels), where each pixel can take on one of 32 intensity values (5 bits/pixel). Assume that 30 pictures are sent per second. Find the source rate R (bps). Assume that the TV picture is to be transmitted over a channel with 4.5-MHz bandwidth and a 35-dB signal-to-noise ratio. Determine the channel capacity (bps).

(8 marks)

bps = (480 × 500) pixels × 5 (bit/pixel) × 30 pictures/s

bps = 480 × 500 × 5 × 30 = $\boxed{36 \text{ Mbps}}$

36×10^6 bps

Capacity = bandwidth $\log_2(1 + SNR)$

= 4.5 MHz × $\log_2(1 + 3162)$

$\boxed{\text{Capacity} = 52.32 \text{ Mbps}}$

= 52321850 bps

$SNR_{dB} = 10 \log_{10}(SNR)$

$35 = 10 \log_{10}(SNR)$

$3.5 = \log_{10}(SNR)$

SNR = (3162.3)

8

GOOD LUCK

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